

DESIGN EVALUATION 1

XR Whiteboard User Testing Guide and Questions

Name	_____	Date	_____
Prototype	XR Study Whiteboard	Round	1

What this evaluation tests

Can users use the XR controllers and visible rays to draw accurately on the wall whiteboard and the paper on the table? It also tests the Pencil, Eraser, Clear buttons, menus, comfort, and confidence.

1. Testing aim and success measures

Use this aim: I want to learn whether users can find a surface, choose a tool, draw three shapes, and understand the controls without help.

Metric	Success target
Find a surface	User reaches the whiteboard and table paper without help.
Draw on both surfaces	User draws a continuous horizontal line, vertical line, and circle on each surface.
Stroke quality	Lines are smooth, continuous, and accurate. No dots, gaps, random marks, or sudden artifacts.
Tool controls	User can choose Pencil, use Eraser, and use Clear without confusion.
XR interaction	Controller rays point at the surface and buttons. User feels in control and safe.
Confidence	User rates the experience 4 or 5 out of 5.

2. Test setup

- Participants: Test [number] target users. Record prior VR experience.
- Setting: Use the classroom scene with the wall whiteboard and a table paper.
- Evidence: Record task success, mistakes, time, quotes, questions, and visible drawing problems.
- Do not lead the user. Give the same instructions to everyone and only help if they are stuck.

3. User tasks

Task	Instruction to the participant	Observe
1. Explore	Look around and find the wall whiteboard and table paper.	Does the user know where to go?
2. Whiteboard	Choose Pencil. Draw a horizontal line, vertical line, and circle.	Accuracy, ray position, smoothness, confidence.
3. Paper	Choose Pencil. Draw the same three shapes on the paper.	Can the user move from wall to table? Any UI or surface confusion?
4. Erase	Choose Eraser and remove part of one shape.	Can the user find and use Eraser?
5. Clear	Use the clear control to reset the paper or board.	Does the user understand the button and confirmation?

4. Questions to ask

- Before testing: Have you used an XR headset or controllers before? What do you expect to do first?
- During testing: What are you trying to do now? What do you expect this button or ray to do?
- During testing: What is confusing, hard to reach, hard to read, or blocking your view?
- After testing: Which surface was easier to use, the whiteboard or paper? Why?
- After testing: Did the line and circle look smooth and continuous? Did you see any dots, gaps, or unwanted marks?
- After testing: Were the controllers, rays, menus, and buttons easy to understand?
- After testing: What is one change that would make this XR whiteboard better?

5. Results: record what happened

Keep this section factual. Do not explain the cause yet. Use numbers and short quotes.

User	Whiteboard result	Paper result	Quote or visible issue
P1	Completed / needed help / failed	Completed / needed help / failed	
P2	Completed / needed help / failed	Completed / needed help / failed	
P3	Completed / needed help / failed	Completed / needed help / failed	
P4	Completed / needed help / failed	Completed / needed help / failed	

Write facts

Example: Four users completed the horizontal line. Two users started drawing before selecting Pencil. Three users asked what Clear Paper would do.

6. Analysis and insights

Group repeated behaviours into themes. Link every insight to evidence from the test.

Pattern or theme	Evidence	Meaning for the XR whiteboard
Users understood...	Users did or said...	This shows...
Users struggled with...	Users did or said...	This may mean...
Users preferred...	Whiteboard or paper...	The design should...
Users noticed...	Ray, menu, stroke, or button...	This affects...

Sentence starter: The repeated behaviour was [pattern]. This matters because [user need or design problem].

7. Check the testing aims

Aim	Status	Evidence and reason
Users can find and use both surfaces.	Validated / invalidated / uncertain	
Users can draw three clean shapes on both surfaces.	Validated / invalidated / uncertain	
Users understand Pencil, Eraser, and Clear.	Validated / invalidated / uncertain	
Controllers, rays, and UI support confident use.	Validated / invalidated / uncertain	

Be honest

An aim can be partly validated. If the evidence is not strong enough, mark it uncertain and state what you need to test next.

8. Concept iteration

Turn the evidence into clear design decisions. State what you will change, keep, or test again.

Insight from testing	Design response	Why this response fits
Users had trouble reaching or reading the menu.	Move, resize, or relabel...	It makes the control...
Strokes showed gaps or unwanted marks.	Improve ray, input, or stroke handling...	It improves...
Users understood a control or surface.	Keep or strengthen...	It supports...
An issue is still uncertain.	Test again with...	It will answer...

Updated concept description

- The XR whiteboard now helps users...
- The main change is... because the evidence showed...
- The next prototype will improve...

9. Reflection and next steps

- What worked well in the test method?
- What did not work, such as task wording, timing, setup, or recording?
- What did you learn about drawing in XR and moving between the two surfaces?
- What will you change in the next prototype?
- What will you test next, and what result would change your decision?

Next question	Method	Evidence needed
I still need to learn...	I will observe / ask / measure...	I will look for...

10. Final checklist

- The report has no more than three content pages.
- Results say what users did, said, and experienced.
- Insights are connected to evidence and user needs.
- Aims are marked validated, invalidated, or uncertain.
- Changes are specific and justified by the XR test.
- The next prototype and next test are clear.
- Raw notes, full quotes, and full data are in an appendix.

Important

Do not only say that the prototype was good or bad. Explain what the user did, what that means, and what you will change because of it.